A COMPLETE RING WHOSE FINITELY GENERATED IDEAL HAS OPEN POWER CLOSURES IS ADIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma, delivered as a navigation stub. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of the identity map from the I-adic topology to the given topology. Same chapter, Definition definition-taut. The hypothesis that the power closures are open and form a fundamental system is recognised as tautness of A' -> A.
- A taut continuous ring map from a pre-adic ring to a complete ring makes the target adic with a finitely generated ideal of definition. Same chapter, Lemma lemma-taut-is-adic, invoked at proof lines 1709-1710. Delivers both conclusions of the lemma.

Source reference. Original lemma 0APU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1557–1557. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a complete linearly topologized ring. Let $I \subset A$ be a finitely generated ideal such that the closure of I^n is open for all $n \geq 0$ and these closures form a fundamental system of open ideals. Then A is adic and has a finitely generated ideal of definition.*

Proof. Denote A' the ring A endowed with the I-adic topology. The assumptions tells us that $A' \to A$ is taut. We conclude by Lemma 0APU (to be sure, this lemma also tells us that I is an ideal of definition). □

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.